

# RIYA PALESHA

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## Education

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| <b>Pune Institute of Computer Technology</b><br><i>BE in Computer Engineering</i>                    | <b>2020 – 2024</b><br>Grade : 9.53       |
| <b>Indian Institute of Technology, Madras</b><br><i>BSc in Programming and Data Science (online)</i> | <b>2020 – 2025</b><br>Grade : 8.70       |
| <b>Samubai Mulchand Choksey High School And Junior College</b><br><i>Stream - Science</i>            | <b>2018 – 2020</b><br>Percentage : 94.62 |

## Experience

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| <b>Barclays</b>   <i>Software Engineer - BA4</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>July 2024 – Present</b>  |
| <ul style="list-style-type: none"><li>Contributed to the disaster recovery region setup for the bank's Fraud Platform on AWS, enhancing system resilience and ensuring service continuity during regional outages.</li><li>Designed and implemented monitoring and alerting solutions using Grafana and Prometheus, improving system observability and reducing issue-detection time.</li><li>Independently executed the complex week-long restacking workflow of Aerospike database across multiple environments.</li><li>Developed and deployed AWS Lambda functions and CloudFormation Templates to automate infrastructure provisioning and optimize workflows across environments.</li><li>Spearheaded the infrastructure setup and access control for a new AWS MSK cluster utilizing IAM authentication; designed scoped roles and custom alerting that enabled application teams to securely deploy isolated operations.</li><li>Automated Aerospike configuration workflows using GitLab CI/CD pipelines, eliminating manual intervention and reducing configuration-related deployment errors by over 70%.</li></ul> |                             |
| <b>Barclays</b>   <i>Summer Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>June 2023 – Aug 2023</b> |
| <ul style="list-style-type: none"><li>Developed automated Jira governance workflows using Groovy scripts and Jenkins pipelines, reducing manual tracking efforts by more than 60%.</li><li>Designed UI components for the Fraud Platform Dashboard using React, enhancing usability and operational visibility for internal teams.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                             |

## Projects

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| <b>ASKIVA: AI-Powered Academic Chatbot</b>   <i>Next.js, Tailwind CSS, Flask, SQLAlchemy, ChromaDB, LangChain</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>March 2025</b>    |
| <ul style="list-style-type: none"><li>Built an AI-enhanced IITM academic portal with automated concept simplification and doubt resolution using LangChain and Google generative models.</li><li>Developed secure backend services with Flask APIs, JWT authentication, SQLAlchemy ORM, and ChromaDB-based vector retrieval.</li><li>Implemented role-based dashboards and a robust Retrieval-Augmented Generation (RAG) pipeline to deliver personalized workflows and accurate, context-aware academic assistance.</li></ul>                                                                                                                 |                      |
| <b>Data-Driven Business Performance Analysis</b>   <i>Business Analytics, Python, Data Visualization, EDA</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>October 2024</b>  |
| <ul style="list-style-type: none"><li>Analyzed real operational, financial, and customer-feedback datasets from a retail business using statistical trends and multi-chart visualizations.</li><li>Identified irregular payment cycles, high-demand SKUs, overstock patterns, and service pain points using turnover metrics and sentiment analysis.</li><li>Proposed data-driven recommendations for payment policies, inventory optimization, and service improvements to enhance operational efficiency.</li></ul>                                                                                                                          |                      |
| <b>Legal Document Understanding &amp; Query Assistant</b>   <i>React, FastAPI, MongoDB, PyPDF2, LangChain</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>January 2024</b>  |
| <ul style="list-style-type: none"><li>Engineered an end-to-end platform to simplify complex legal contracts, implementing large-document text extraction, chunking techniques, clause segmentation, and interactive contract navigation.</li><li>Developed semantic search and question-answering features using NLP/LLM-inspired retrieval techniques, and built MongoDB-backed version comparison to detect and highlight document changes.</li><li>Designed resilient pipelines to handle noisy real-world PDFs and domain-specific terminology; co-authored a survey paper on contemporary methods in legal-document processing.</li></ul> |                      |
| <b>Automated Performance Analysis of Shortest Path Algorithms on Directed Graphs</b>   <i>C++</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>November 2022</b> |
| <ul style="list-style-type: none"><li>Compared the performance of Dijkstra's, Bellman-Ford, and Floyd-Warshall across varying graph sizes and weight distributions to study scalability and efficiency.</li><li>Designed modular components for graph construction, algorithm execution, timing, and result logging, ensuring repeatable and extensible experiments.</li></ul>                                                                                                                                                                                                                                                                 |                      |

## Research Work

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### Reducing Hallucinations through Enhanced Retrieval-Augmented Generation Pipelines

- Built a custom RAG pipeline using Ollama, and FAISS/ChromaDB to study techniques for reducing LLM hallucinations.
- Implemented semantic search, reranking, chunking strategies, query reformulation, and confidence scoring to improve factual grounding.
- Conducted experiments across multiple document formats (i.e. DOC,DOCX,TEXT,CSV,PDF) developing 250+ queries and achieving 94% accuracy with under 2% hallucination.

## Technical Skills

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**Programming Languages :** Python, C/C++, Java, JavaScript, SQL, HTML/CSS

**Frameworks & Libraries :** React, Next.js, Vue.js, Node.js, Flask, FastAPI, PyTorch, Hugging Face, NumPy, Pandas, LangChain

**Databases :** PostgreSQL, MongoDB, Aerospike

**Cloud & DevOps :** AWS, Jenkins, Kafka

**Developer Tools :** Git, GitHub, Jira, Postman, Insomnia, VS Code, Eclipse

**Core Concepts:** Data Structures and Algorithms, Operating Systems, Databases, Computer Networks, Distributed Systems, Machine Learning, Artificial Intelligence

## Leadership / Extracurricular

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- PCSB (PICT CSI Student Branch): Organized and led technical workshops and events for 100+ students, fostering peer learning and collaboration.
- Developed GenAI-based prototypes during Barclays Accelerate and ReImagine Hackathons to automate compliant CloudFormation Template generation and fraud-detection rule creation.
- Rotaract Club of Pune Phoenix: Contributed to community service initiatives focused on education, health, and youth development.
- Awarded **Best Capstone Project** for developing a Library Management System at IIT Madras.
- Secured **1st rank** among 300+ students in SSC Board examinations.